

SALES & SUPPLY CHAIN

SQL Exploratory Data Analysis

Visual Business Case Study

From transactional data to commercial and operational insights

Tools: MySQL • SQL EDA • Relational Joins • Aggregation • Window Functions

1. Executive Summary

Objective

Use SQL to analyze a sales and supply-chain dataset across orders, customers, products, order items, and shipments.

What was analyzed

Commercial performance was examined alongside operational performance: revenue, products, customers, sales channels, delivery outcomes, delay rates, regional patterns, customer types, and transportation costs.

Main takeaway

The project shows why sales performance should be considered together with supply-chain execution. Revenue concentration and delivery reliability vary across customers, channels, regions, product categories, and customer types.

Portfolio value

The analysis demonstrates practical SQL skills while translating transactional data into business-oriented findings and recommendations.

2. Data Model & Analytical Approach

Core tables

`ssales\_order2` contains order-level information; `customers` contains customer and regional attributes; `products` contains product/category information; `sales\_order\_items` contains quantities and revenue; `shipments` contains delivery and transportation information.

Key relationships

Orders connect to customers through `customer_id`. Orders connect to order items through `order_id`. Order items connect to products through `product_id`. Shipments connect back to orders through `order_id`.

Why joins matter

The analysis combines these tables so that transactional revenue can be interpreted by customer, product, region, channel, and logistics outcomes rather than analyzed in isolation.

3. Data Cleaning & Quality Checks

Duplicate detection

Duplicate orders were investigated using grouped checks and ROW_NUMBER(). A cleaned order table was created and duplicate rows were removed based on the generated row number.

Missing values

Important order fields were checked for NULL values, including order ID, customer ID, order date, requested delivery date, order status, and sales channel.

Referential integrity

Orders were checked against the customer table to identify customer IDs that did not exist in the customer master.

Date validation

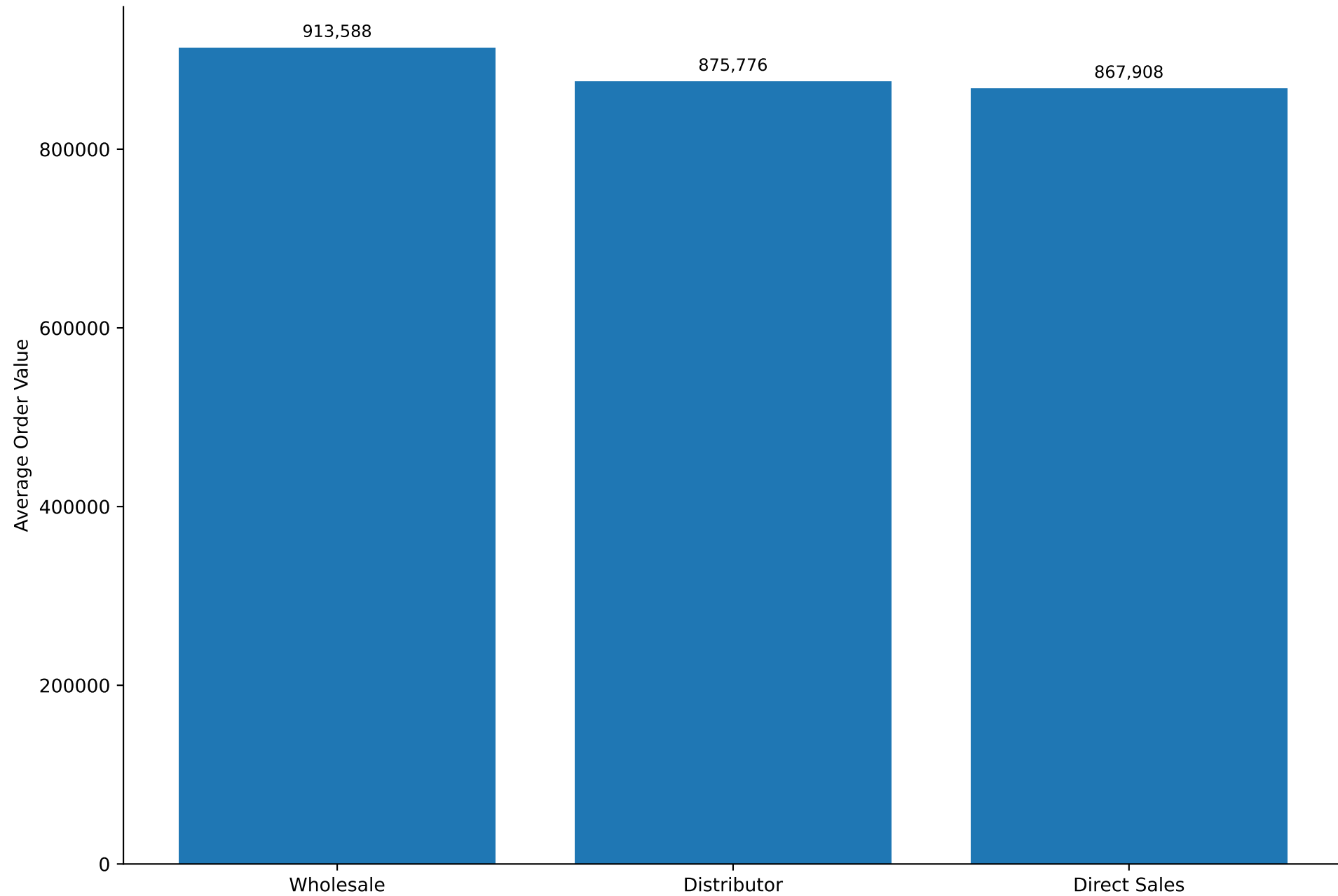
Orders were checked for cases where the order date occurred after the requested delivery date.

Schema cleanup

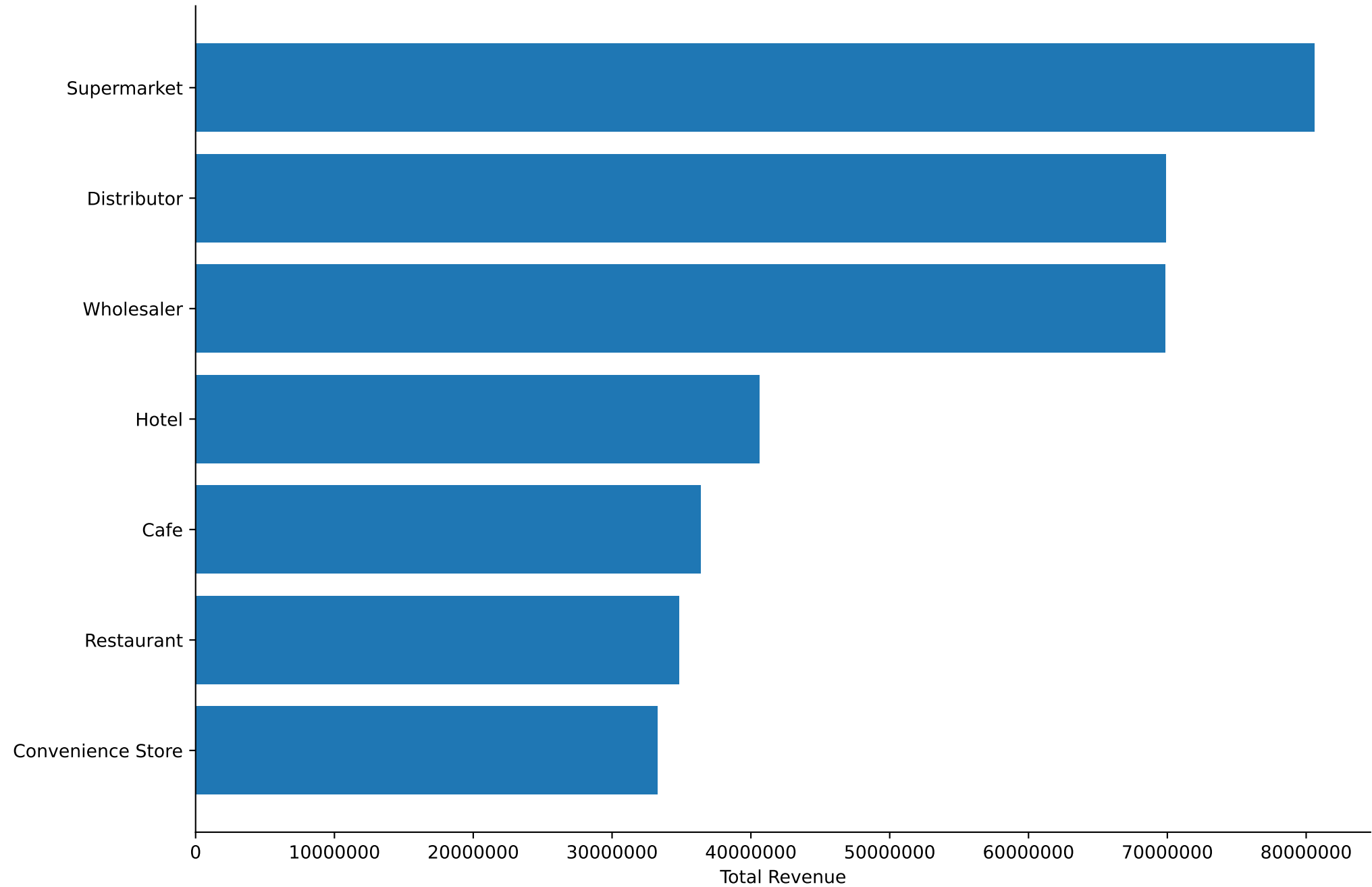
Imported column names containing encoding artifacts such as `ï»¿` were renamed to clean field names.

4. Sales Channel Performance

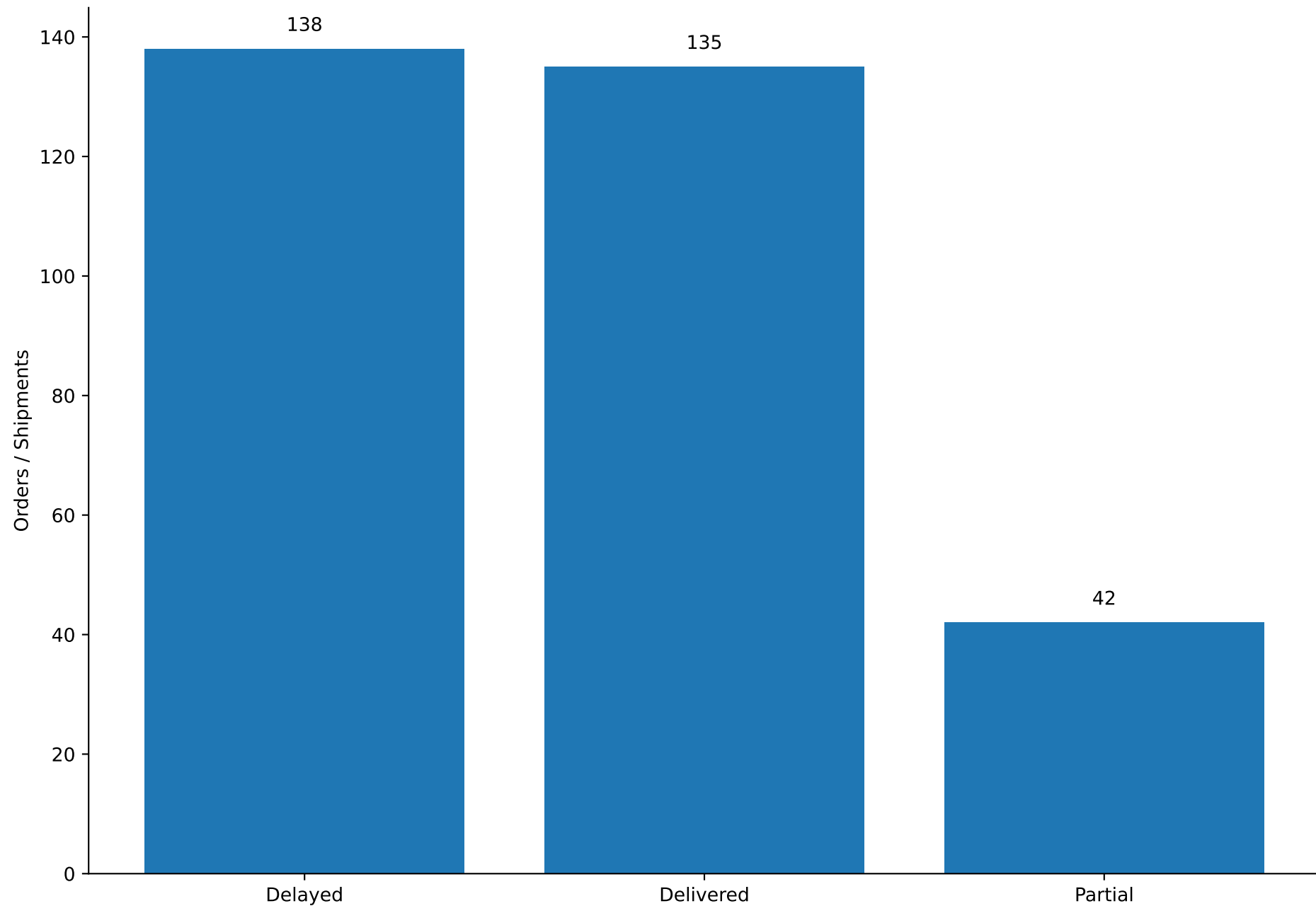
Average order value from the analyzed main sales channels



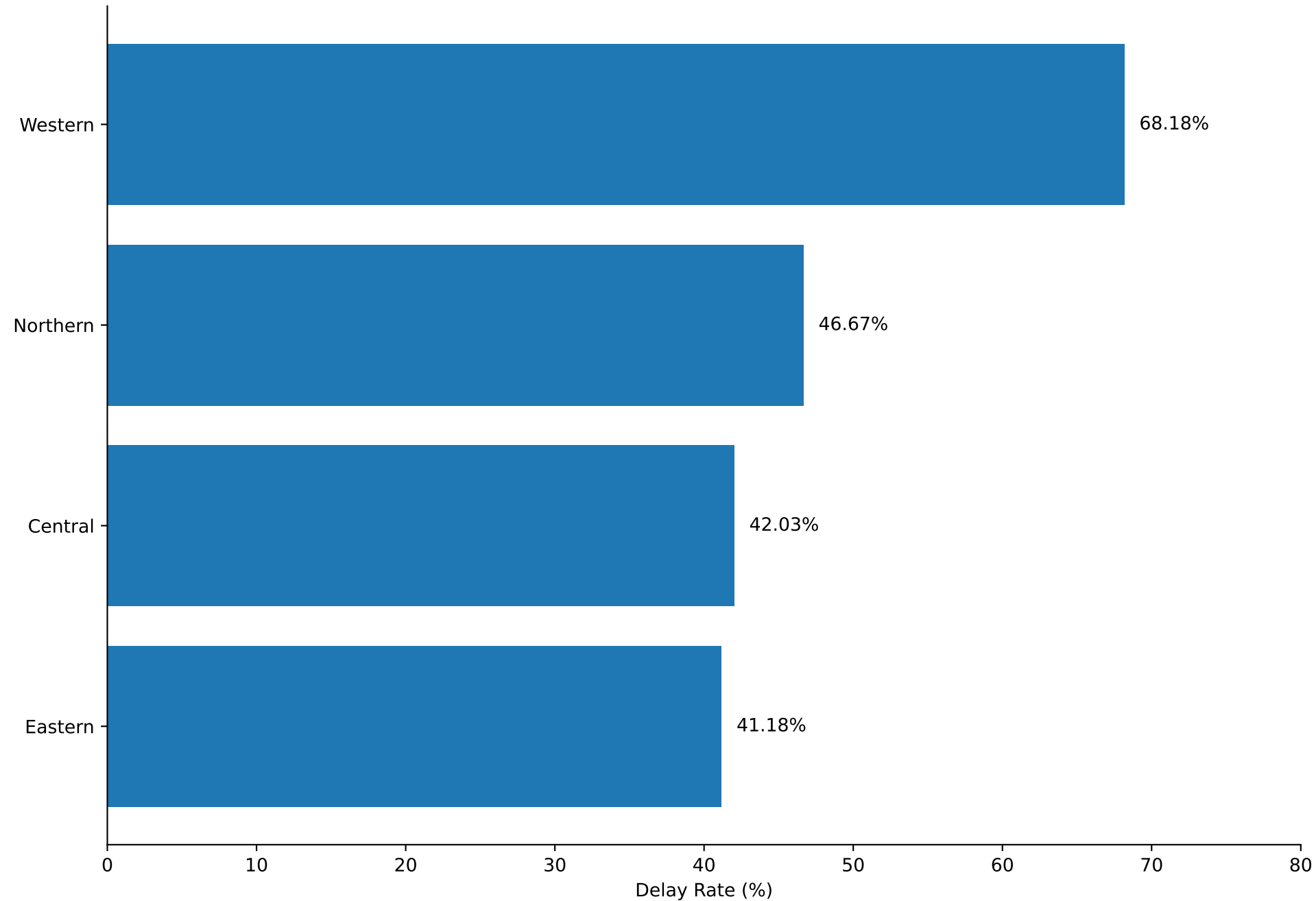
5. Revenue by Customer Type



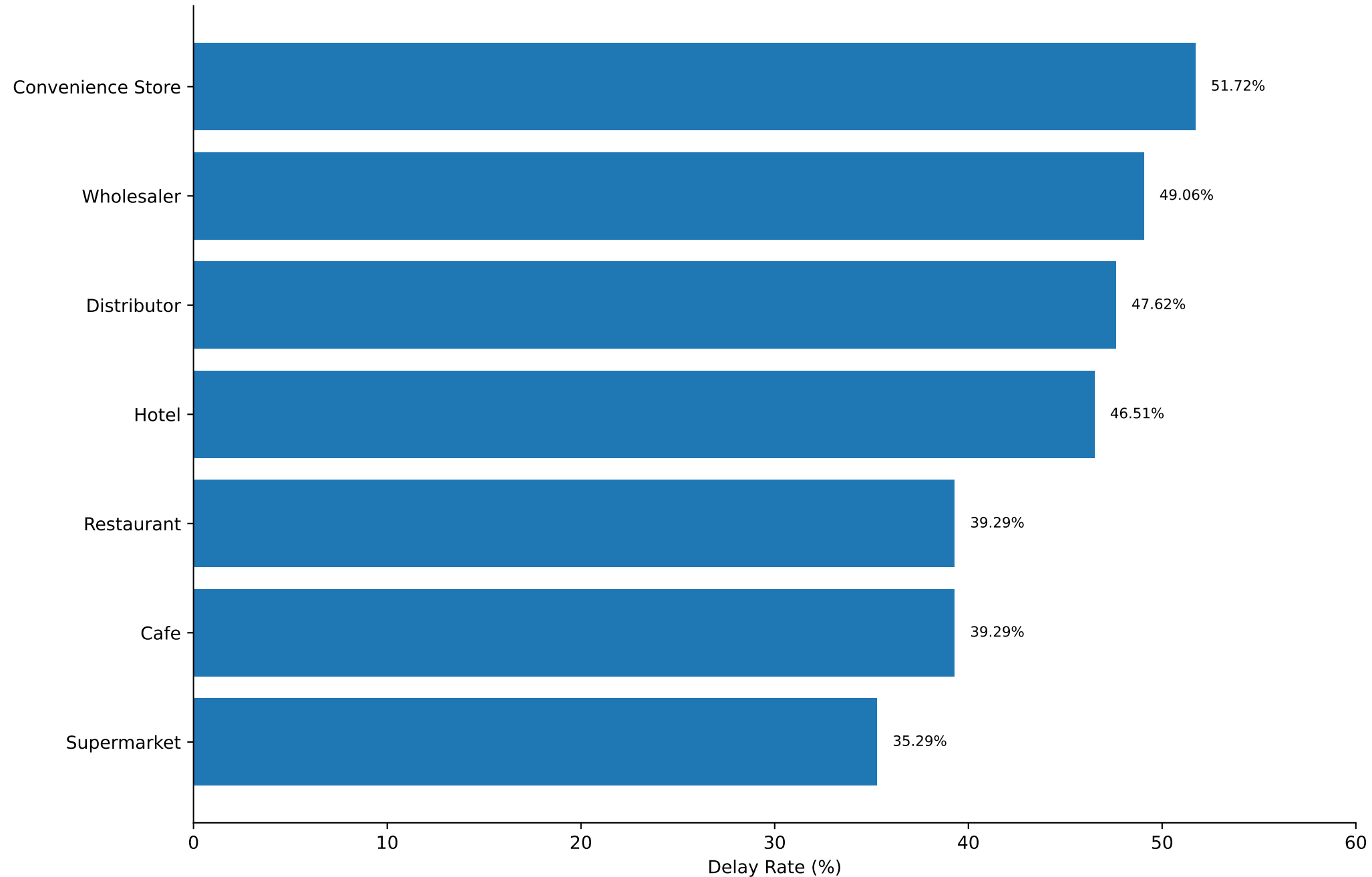
6. Shipment Delivery Outcomes



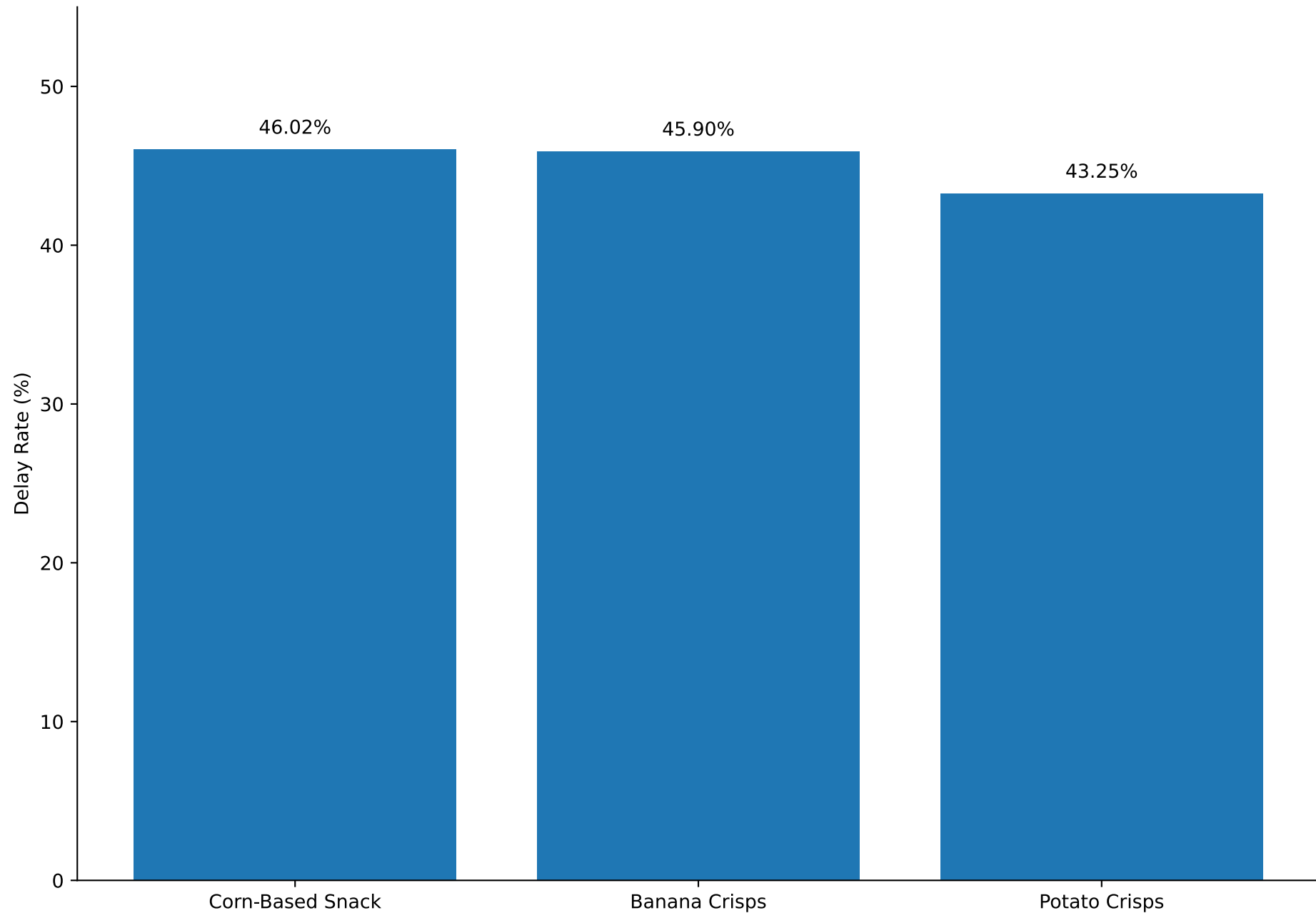
7. Delivery Delay Rate by Region



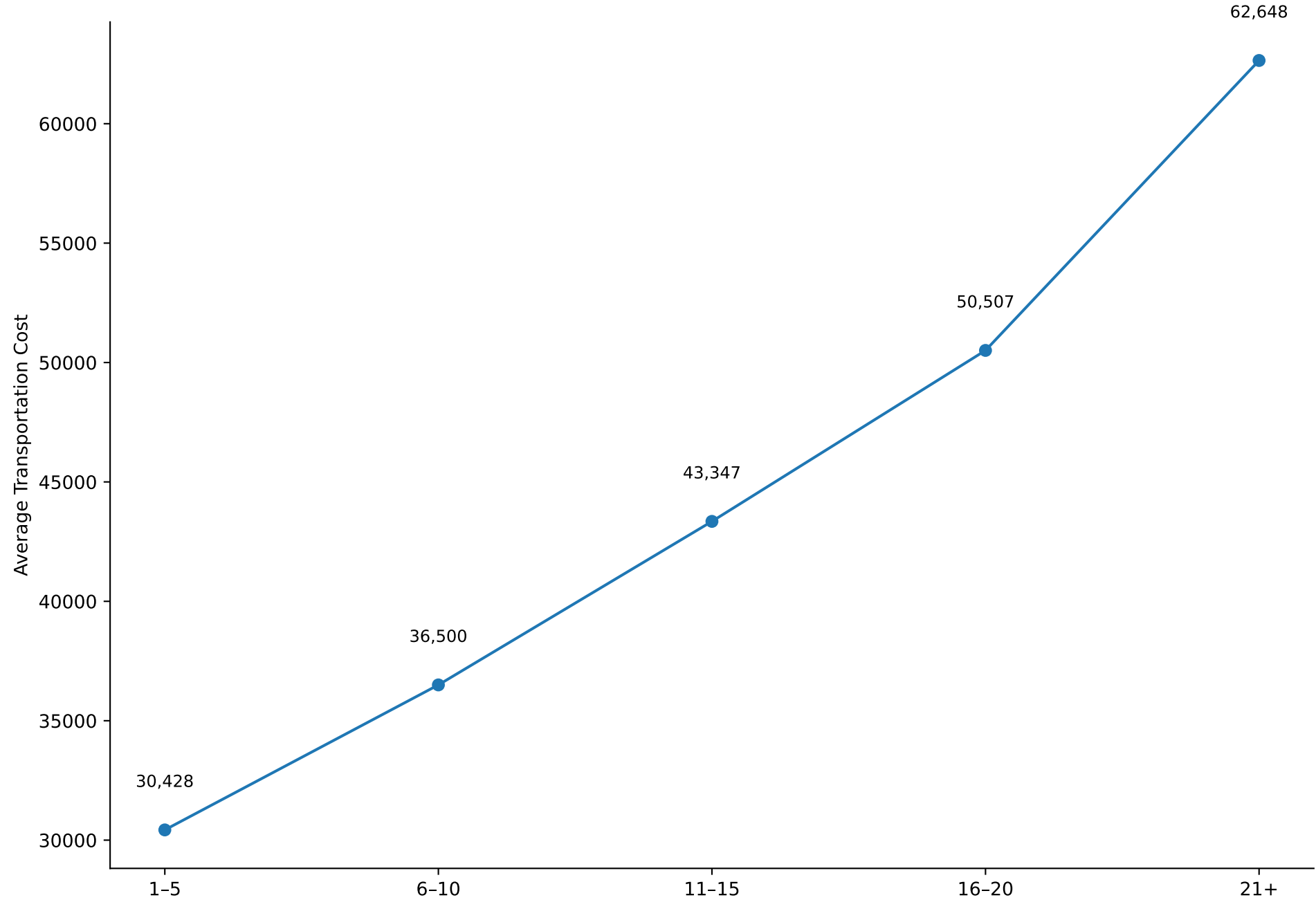
8. Delivery Delay Rate by Customer Type



9. Delivery Delay Rate by Product Category



10. Transportation Cost by Shipment Size



11. Key Findings

Sales channels

Wholesale showed the highest average order value among the three main channels at approximately 913,588, followed by Distributor at approximately 875,776 and Direct Sales at approximately 867,908.

Customer types

Supermarkets generated the highest total revenue among customer types in the analyzed results. Restaurants and Cafes had higher average order values than several other customer groups.

Delivery

The shipment-status result contained 138 Delayed, 135 Delivered, and 42 Partial shipments, showing that delivery execution is a material part of the dataset.

Regional reliability

Western had the highest observed regional delay rate at 68.18%, followed by Northern at 46.67%, Central at 42.03%, and Eastern at 41.18%.

Customer-type reliability

Convenience Stores had the highest delay rate at 51.72%, while Supermarkets had the lowest among the listed customer types at 35.29%.

Product categories

Corn-Based Snacks had the highest category delay rate at 46.02%, followed closely by Banana Crisps at 45.90% and Potato Crisps at 43.25%.

Transportation

Average transportation cost increased steadily as shipment size increased, from 30,428 for 1-5 cases to 62,648 for 21+ cases.

12. Business Recommendations

1 — Prioritize high-delay segments

Investigate the highest-delay regions, customer types, channels, and product categories for route, fulfillment, carrier, or process constraints.

2 — Protect high-value customers

Use average order value alongside revenue to identify customers that may warrant differentiated account-management and service strategies.

3 — Review lower-performing products

Products with lower revenue should be evaluated for demand, pricing, distribution, and inventory implications before making portfolio decisions.

4 — Examine shipment economics

Because transportation cost rises with shipment size, evaluate whether shipment consolidation, routing, delivery frequency, or other logistics decisions could improve efficiency.

5 — Connect commercial and operational KPIs

Revenue and delivery performance should be considered together. A strong sales segment may still require operational improvement if its delay rate is high.

13. SQL Skills Demonstrated

Querying & aggregation

SELECT, WHERE, GROUP BY, ORDER BY, HAVING, COUNT, COUNT(DISTINCT), SUM, AVG, and ROUND.

Conditional analysis

CASE WHEN was used to create analytical groups and calculate conditional metrics such as delayed-order counts and delay rates.

Relational analysis

JOIN and LEFT JOIN were used to combine orders, customers, products, order items, and shipments.

Advanced SQL

CTEs, subqueries, ROW_NUMBER(), date functions, and conditional aggregation were used throughout the analysis.

Data preparation

CREATE TABLE, INSERT INTO, DELETE, and ALTER TABLE were used during cleaning and schema preparation.

14. Conclusion

Overall

This project demonstrates an end-to-end SQL EDA workflow that moves from raw transactional data through data-quality checks, relational joins, descriptive analysis, operational analysis, and business recommendations.

Portfolio takeaway

The strongest aspect of the project is the connection between commercial and supply-chain performance. The analysis does not stop at revenue rankings; it investigates where delivery reliability and transportation economics may affect business performance.

Final deliverables

The portfolio package consists of the organized SQL script, README documentation, and this visual case-study report.